Gaussian quadrature using Hermite polynomials

Gauss-Hermite quadrature

- **Gauss-Hermite quadrature** is used to approximate integrals over the whole real line with a Gaussian weight

$$\int_{-\infty}^{\infty} f(x)e^{-x^2}\,dx \approx \sum_{i=1}^{n} w_i f(x_i)$$

x_i are the roots of the physicists' Hermite polynomial $H_n(x)$,

w_i are the corresponding quadrature weights,

the rule is exact for all polynomials $f(x)$ of degree up to $2n-1$.

Gauss-Hermite quadrature

- The weights are

$$w_i = \frac{2^{n-1} n! \sqrt{\pi}}{n^2 \left[H_{n-1}(x_i) \right]^2}$$

- x_i are the roots of $H_n(x)$

Example: 2-point Gauss-Hermite quadrature

- For n=2, $$H_2(x) = 4x^2 - 2.$$
- The roots are $$x_1 = -\frac{1}{\sqrt{2}}, \qquad x_2 = \frac{1}{\sqrt{2}}.$$
- The weights are $$w_1 = w_2 = \frac{\sqrt{\pi}}{2}.$$
- This is exact for polynomials up to degree 3

$$\int_{-\infty}^{\infty} f(x)e^{-x^2}\,dx \approx \frac{\sqrt{\pi}}{2}\left[f\left(-\frac{1}{\sqrt{2}}\right) + f\left(\frac{1}{\sqrt{2}}\right)\right]$$

Example: 3-point Gauss-Hermite quadrature

- For n=3,

$$H_3(x) = 8x^3 - 12x.$$

- The roots are

$$x_1 = -\sqrt{\frac{3}{2}}, \qquad x_2 = 0, \qquad x_3 = \sqrt{\frac{3}{2}}.$$

- The weights are

$$w_1 = w_3 = \frac{\sqrt{\pi}}{6}, \qquad w_2 = \frac{2\sqrt{\pi}}{3}.$$

- This is exact for polynomials up to degree 5

$$\int_{-\infty}^{\infty} f(x)e^{-x^2}\,dx \approx \frac{\sqrt{\pi}}{6} f\left(-\sqrt{\frac{3}{2}}\right) + \frac{2\sqrt{\pi}}{3} f(0) + \frac{\sqrt{\pi}}{6} f\left(\sqrt{\frac{3}{2}}\right)$$

Laplace approximation is essentially 1-point adaptive Gauss-Hermite quadrature

- Gauss-Hermite quadrature and the Laplace approximation are closely related because both approximate integrals by exploiting a Gaussian approximation around the dominant point of the integrand.
- Laplace approximation uses only the local Gaussian approximation around the mode.
- Adaptive Gauss-Hermite quadrature does the same centering and scaling, but then improves the approximation by evaluating the integrand at multiple points around the mode.

Laplace approximation

- Suppose we want to approximate an integral of the form

$$I = \int_{-\infty}^{\infty} e^{\ell(x)}\, dx,$$

- Define

$$\hat{x} = \arg\max_x \ell(x) \qquad\qquad H = -\ell''(\hat{x}) > 0.$$

- Near the maximum, a second-order Taylor expansion gives

$$\ell(x) \approx \ell(\hat{x}) - \frac{1}{2} H (x - \hat{x})^2.$$

Laplace approximation

- Using the quadratic approximation

$$I = \int e^{\ell(x)}\, dx \approx e^{\ell(\hat{x})} \int e^{-\frac{1}{2}H(x-\hat{x})^2}\, dx.$$

- Therefore,

$$I \approx e^{\ell(\hat{x})} \sqrt{\frac{2\pi}{H}}$$

- It replaces the whole integrand by a Gaussian centered at the mode.

Adaptive Gauss-Hermite quadrature

- Gauss-Hermite quadrature approximates integrals of the form

$$\int_{-\infty}^{\infty} f(z)e^{-z^2}\,dz \approx \sum_{i=1}^{n} w_i f(z_i),$$

- To connect this with Laplace approximation, make the change of variables

$$x = \hat{x} + \sqrt{\frac{2}{H}}z.$$

Adaptive Gauss-Hermite quadrature

- Factor out the Gaussian weight

$$I = \int e^{\ell(x)}\, dx = \sqrt{\frac{2}{H}} \int_{-\infty}^{\infty} e^{\ell\left(\hat{x}+\sqrt{\frac{2}{H}}z\right)}\, dz.$$

- This is adaptive Gauss-Hermite quadrature because the nodes are shifted to the mode and scaled by the local curvature H

$$I = \sqrt{\frac{2}{H}} \int_{-\infty}^{\infty} e^{-z^2} \left[e^{z^2+\ell\left(\hat{x}+\sqrt{\frac{2}{H}}z\right)}\right] dz.$$

Laplace as one-point Gauss-Hermite quadrature

- For one-point Gauss-Hermite quadrature,

$$z_1 = 0, \qquad w_1 = \sqrt{\pi}.$$

- Substituting into the adaptive Gauss-Hermite formula gives

$$I \approx \sqrt{\frac{2}{H}}\sqrt{\pi}\exp\left[0 + \ell(\hat{x})\right].$$

$$\approx e^{\ell(\hat{x})}\sqrt{\frac{2\pi}{H}}.$$